

Robust Offline Safe RL under Dynamic Safety Budgets

From CCAC to Soft & Calibrated OOD Regularization

RL Mid-term Presentation

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Background: Offline Safe RL

Traditional RL learns by repeatedly interacting with the environment.

In safety-critical domains, online exploration can be too expensive or risky, so we want to learn from an offline dataset.

Robotics

Avoid collisions, falls, and unsafe motions

Autonomous Driving

Unsafe exploration cannot be used to collect data

Healthcare

Policies must satisfy strict risk budgets

Problem Setup

Offline Safe RL optimizes reward while constraining cumulative safety cost:

$$\max_{\pi} \mathbb{E}_{\pi}[R(\tau)] \quad \text{s.t.} \quad \mathbb{E}_{\pi}[C(\tau)] \leq \epsilon$$

It must solve three problems at the same time:

1. OOD state-action pairs caused by offline data
2. A policy that satisfies the cost constraint
3. A safety budget that may change during deployment

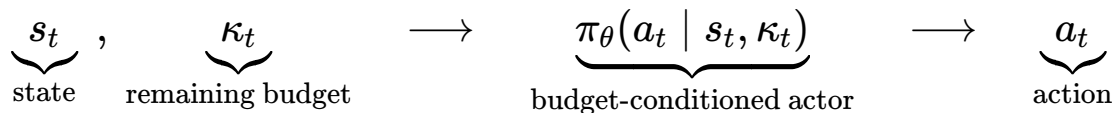
Goal: learn a high-reward and safe policy that can adapt to different remaining budgets.

CCAC: Existing Backbone

CCAC addresses this setting with a budget-conditioned actor-critic framework.

$$\pi_{\theta}(a_t \mid s_t, \kappa_t), \quad \kappa_{t+1} = \kappa_t - c_t$$

The remaining budget κ_t becomes part of the policy input, so the same policy can behave differently under different safety budgets.



Take-away: CCAC turns a fixed-threshold safe RL problem into a dynamic-budget policy learning problem.

CCAC: How Safety Is Enforced

The key mechanism is to identify generated OOD / unsafe state-action pairs and penalize them through the cost critic.

Module	Role	Notation
Offline data	Build budget-annotated transitions	$D = \{(s_t, a_t, r_t, c_t, s_{t+1}, \kappa_t)\}$
Constraint-conditioned CVAE	Generate state-action pairs under a budget	$(\hat{s}, \hat{a}) \sim p_\phi(s, a \mid z, \kappa)$
OOD classifier	Decide whether (s, a, κ) is OOD / unsafe	$h_\psi(s, a \mid \kappa) > 0.5$
Cost critic	Overestimate cost for unsafe samples	$Q_c(s, a \mid \kappa)$
Actor	Maximize reward while avoiding high cost	$\pi_\theta(a \mid s, \kappa)$

$$p_\phi(s, a \mid z, \kappa) \rightarrow h_\psi(s, a \mid \kappa) \rightarrow Q_c(s, a \mid \kappa) \rightarrow \pi_\theta(a \mid s, \kappa)$$

Our Observation: A Brittle OOD Gate

CCAC relies heavily on the OOD classifier. If classifier decisions are inaccurate, cost overestimation can become unstable.

The original hard OOD rule is:

$$\mathbb{I}_{\text{OOD}}(s, a, \kappa) = \mathbb{I}[h_{\psi}(s, a \mid \kappa) > 0.5]$$

This can cause two issues:

1. Classifier scores may not be calibrated; 0.49 and 0.51 are treated completely differently
2. Under imbalanced data, the classifier may miss truly dangerous OOD samples

Question: can we make OOD regularization softer and more calibrated?

Our Small Improvement

Soft & Calibrated OOD Regularization

Instead of hard filtering, use a soft OOD weight:

$$w_{\text{OOD}}(s, a, \kappa) = \sigma(\alpha(h_{\psi}(s, a \mid \kappa) - \tau))$$

Then use w_{OOD} to weight the cost overestimation penalty, rather than making a binary decision.

We also improve classifier training with:

- weighted BCE or focal loss for class imbalance
- calibration analysis on classifier scores
- false negative rate analysis for unsafe OOD samples

Expected Benefit and Evaluation

The improvement is small enough for a course project, but directly targets CCAC's safety bottleneck.

Question	What we will measure
Does soft weighting reduce threshold brittleness?	Reward-cost trade-off near the safety boundary
Does calibration improve safety detection?	ECE / reliability curve of OOD scores
Does it catch dangerous OOD samples?	OOD false negative rate
Does budget adaptation remain stable?	Performance under different initial budgets κ_1

Expected outcome: similar or better reward, lower constraint violation, and more stable behavior when the OOD classifier is uncertain.

Feasibility

We plan to build on CCAC and evaluate the modification on DSRL-style benchmark tasks.

Component	Plan
Environments	Run / Circle / Velocity-style safety control tasks
Baseline	Original CCAC with hard OOD threshold
Variant	CCAC + soft OOD weight + calibrated classifier loss
Metrics	Reward, cost, constraint violation, OOD FNR, calibration, budget adaptation
Ablations	hard vs. soft OOD, BCE vs. focal loss, fixed vs. varying budgets

The change is localized to classifier training and cost regularization, so it is feasible without redesigning the whole algorithm.

Related Work

Direction	Methods	What they address	Remaining gap
Offline RL	CQL, IQL, BCQ, BEAR	Distribution shift	No explicit safety constraints
Safe RL	Lagrangian, CPO, PID-Lagrangian	Constrained optimization	Often needs online interaction
Offline Safe RL	CPQ, COptiDICE, VOCE, FISOR	Reward-cost trade-off from offline data	Hard to adapt to changing budgets
Budget-conditioned OSRL	CCAC	Dynamic budget adaptation + OOD detection	Classifier threshold may be brittle

Our angle: make the OOD safety signal softer, calibrated, and easier to evaluate.

Q&A

Thank you!

References

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